

Channel-Code Detection by a Third-Party Receiver via the Likelihood Ratio Test

Arti D. Yardi, Animesh Kumar, and Saravanan Vijayakumaran

Department of Electrical Engineering

Indian Institute of Technology Bombay

Mumbai 400076, India

Email: {arti,animesh,sarva}@ee.iitb.ac.in

Abstract—Channel codebook detection is of interest in cognitive paradigm or security applications. A binary hypothesis testing problem is considered, where a receiver has to detect the channel-code from two possible choices upon observing noise-affected codewords through a communication channel. For analytical tractability, it is assumed that the two channel-codes are linear block codes with identical block-length. In a first, this work studies the likelihood ratio test for minimizing the error probability in this detection problem. In an asymptotic setting, where a large number of noise-affected codewords are available for detection, the Chernoff information characterizes the error probability. A lower bound on the Chernoff information, based on the parameters of the two hypothesis, is established. Further, it is shown that if likelihood based efficient (generalized distributive law or BCJR) bit-decoding algorithms are available for the two codes, then the likelihood ratio test for the code-detection problem can be performed in a computationally feasible manner.

I. INTRODUCTION

A typical modern communication system includes a sender, a receiver, and a channel-code (codebook). The knowledge of channel-code is essential for successful decoding of the transmitted message [1]. It may be of interest for a third-party observer or eavesdropper to “learn” the channel-code used by a sender and a receiver in a communication system. In the cognitive radio paradigm, knowledge of the primary’s codebook is helpful to a secondary (observer) in the overlay scenario [2]. The key idea is that if the secondary (unlicensed user) can decode the message sent by the primary (licensed user), it can “cancel” its effect in the secondary transmission. On the other hand, in security or defense applications, an eavesdropper may want to know the channel-coding scheme used by the sender and receiver to decode the messages.

This problem of learning the channel-code from noisy encoded bits is, generally speaking, difficult since the channel-code can be arbitrary, the channel-code parameters may not be known, and there is noise. For these reasons, a simpler version of the larger problem is considered. For an analytical treatment, we consider the simplest possible *code-detection problem*—noise-affected channel-encoded bits are available to an observer and it has to choose a channel-code from two hypothesis to minimize an error metric of choice. The two hypothesis are from the set of linear block codes, and their block-length is same and known. The probability of detection error, with equally likely hypotheses, is the error-metric. Using

the likelihood-ratio test, it is known that the correct hypothesis is selected with an exponentially decreasing error probability and the exponent is given by Chernoff information [3].

For the code-detection problem with the two hypothesis from the set of linear block codes having equal block-lengths, this paper has the following main contributions:

- 1) It is shown that the Chernoff information for the code-detection problem is strictly non-negative if the two hypothesis are different. A non-zero lower bound on the Chernoff information is established to show this result.
- 2) The likelihood ratio test is optimal, but likelihood computation can be difficult. Two methods to compute the likelihood ratio are detailed which bank upon existing efficient decoding algorithms. It is shown that if decoding is computationally efficient via generalized distributive law [4] or BCJR algorithms [5], then likelihood ratio for the code-detection problem can be computed efficiently.

Organization: The system model is explain in Sec. II. The likelihood ratio test and associated Chernoff information lower bounds are discussed in Sec. III. This section contains the main result. Using the BCJR algorithm or the generalized distributive law, efficient methods to compute the likelihood for code-detection are presented in Sec. IV. Performance results for the likelihood test are presented in Sec. V. Conclusions and future work are discussed in Sec. VII.

Related work: The code-detection problem with convolutional codes as hypotheses was considered by Moosavi and Larsson [6]. Their method relies on single parity check equation. Using single parity check equation noise-affected coded bits or random noise has been distinguished by Chabot [7]. This problem has also been approached using the generalized likelihood ratio test by Yardi and Vijayakumaran [8]. In contrast, we apply the likelihood ratio test on the code-detection problem to *minimize* the error probability. In a different setup, reconstruction (estimation) of channel-codes from noise affected channel-encoded bits has also been studied [9], [10], [11].

II. SYSTEM MODEL AND ASSUMPTIONS

A third-party gets access to a sequence of (channel) encoded message bits being transmitted by a sender to a receiver. The transmitter has two possible choices for error-control code and

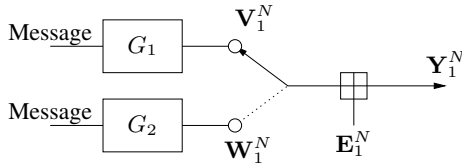


Fig. 1. The model for the proposed code-detection problem is illustrated. Equally likely messages $\mathbf{V}_1^N = [\mathbf{V}_1, \dots, \mathbf{V}_N]$ or $\mathbf{W}_1^N = [\mathbf{W}_1, \dots, \mathbf{W}_N]$ are observed through a binary symmetric channel. The receiver has to detect whether G_1 or G_2 was used for encoding.

the choice made is not known to the third-party. The third-party has to detect which error-control code is being used. The system model is illustrated in Fig. 1. It is assumed that the available choices for error-control are *binary* linear-block codes $\mathcal{C}_1$ and $\mathcal{C}_2$. For a first exposition, the codes $\mathcal{C}_1$ and $\mathcal{C}_2$ are assumed to have *identical block-length* equal to n . Let their generator matrices be G_1 and G_2 , while their parameters be $[n, k_1, d_1]$ and $[n, k_2, d_2]$, respectively. Let $\mathbf{v}_1, \dots, \mathbf{v}_{2^{k_1}}$ be the codewords of code $\mathcal{C}_1$ and $\mathbf{w}_1, \dots, \mathbf{w}_{2^{k_2}}$ be codewords of code $\mathcal{C}_2$. Each transmitted bit, by $\mathcal{C}_1$ or $\mathcal{C}_2$, gets affected by a binary symmetric channel with error probability p . This is shown by adding binary noise vectors $\mathbf{e}_1, \dots, \mathbf{e}_N$ (see Fig. 1). It is assumed that the third-party has noise-affected bit-vectors $\mathbf{y}$, where $\mathbf{y} = \mathbf{v} + \mathbf{e}$ if $\mathcal{C}_1$ was used and $\mathbf{y} = \mathbf{w} + \mathbf{e}$ if $\mathcal{C}_2$ was used by the sender. It is assumed that the received vectors $\mathbf{y}_1, \dots, \mathbf{y}_N$ are synchronized, that is, the start of each received vector is known. The setup is binary and $\mathbf{v}, \mathbf{w}, \mathbf{y}, \mathbf{e} \in \mathbb{F}^n$.

For code-detection (hypothesis testing), it is assumed that a sequence of encoded bit-vectors are available to the third-party. That is, N blocks of received vector $\mathbf{y}$ are available for code-detection. The source bits $\mathbf{V}_1, \dots, \mathbf{V}_N$ is assumed to be independent and identically distributed (i.i.d.), with a uniform distribution over the code $\mathcal{C}_1$. Similarly, $\mathbf{W}_1, \dots, \mathbf{W}_N$ are i.i.d. with a uniform distribution over the code $\mathcal{C}_2$. The channel errors $\mathbf{E}_i, 1 \leq i \leq nN$ are i.i.d. Bernoulli(p) with $p < 1/2$. It follows that $\mathbf{Y}_1, \dots, \mathbf{Y}_N$ are i.i.d. as well.

Thus, the code-detection problem has been formulated as a hypothesis testing problem where the two hypotheses are

$$\begin{aligned} H_1 : \mathbf{Y}_i &= \mathbf{V}_i + \mathbf{E}_i \quad \text{with } \mathbf{V}_i \in \mathcal{C}_1 \\ \text{and } H_2 : \mathbf{Y}_i &= \mathbf{W}_i + \mathbf{E}_i \quad \text{with } \mathbf{W}_i \in \mathcal{C}_2 \end{aligned}$$

for $i = 1, \dots, N$. The two hypotheses are equiprobable. After observing $\mathbf{Y}_1, \dots, \mathbf{Y}_N$ in an i.i.d. manner as described above, a decision H_1 or H_2 has to be made. A sequence of vectors $\mathbf{Y}_1, \dots, \mathbf{Y}_N$ will be denoted as $\mathbf{Y}_1^N$ for any integer N .

III. LIKELIHOOD-RATIO TEST IN CODE-DETECTION

In statistical testing with i.i.d. observations the detection error is characterized by an optimal error-exponent [3, Ch. 12]. In particular, the Chernoff information plays a central role in i.i.d. observation based hypothesis testing problems. The two hypotheses are equally likely by assumption. For the Bayesian hypothesis testing between two distributions, the best achievable error-exponent is given by the Chernoff information

between two distributions [3]. In this section, the Chernoff information for the code-detection problem will be examined.

A. The likelihood-ratio test and likelihood computation

Let $f_{H_j}(\mathbf{y}_1^N) := \mathbb{P}(\mathbf{Y}_1^N = \mathbf{y}_1^N; H_j)$ denote the likelihood of receiving the $\mathbf{y}_1^N$ under the hypothesis H_j , for $j = 1, 2$. Let $l(\mathbf{Y}_1^N)$ be the likelihood ratio. The likelihood-ratio test is [12]

$$l(\mathbf{Y}_1^N) = \frac{f_{H_1}(\mathbf{Y}_1^N)}{f_{H_2}(\mathbf{Y}_1^N)} \underset{H_2}{\overset{H_1}{\gtrless}} \tau, \quad (1)$$

where τ is a pre-determined threshold. Since H_1 and H_2 are equally likely, $\tau = 1$ minimizes the decision error-probability [12]. In our setup, $\mathbf{Y}_1, \dots, \mathbf{Y}_N$ are i.i.d. under the two hypotheses. The likelihood test in (1) simplifies to

$$l(\mathbf{Y}_1^N) = \prod_{i=1}^N \frac{f_{H_1}(\mathbf{Y}_i)}{f_{H_2}(\mathbf{Y}_i)} \underset{H_2}{\overset{H_1}{\gtrless}} 1 \quad (2)$$

To analyze (2), an expression for $f_{H_1}(\mathbf{Y})$ is computed first. An expression for $f_{H_1}(\mathbf{Y})$ is obtained by the total probability law, by conditioning on all possible transmitted codewords,

$$\begin{aligned} f_{H_1}(\mathbf{Y}) &= \sum_{\mathbf{v}_i \in \mathcal{C}_1} f_{H_1}(\mathbf{Y} | \mathbf{V} = \mathbf{v}_i) \mathbb{P}(\mathbf{V} = \mathbf{v}_i; H_1) \\ &= \frac{1}{2^{k_1}} \sum_{\mathbf{v}_i \in \mathcal{C}_1} p^{\text{wt}(\mathbf{Y} + \mathbf{v}_i)} (1-p)^{n - \text{wt}(\mathbf{Y} + \mathbf{v}_i)} \end{aligned} \quad (3)$$

where, $\text{wt}(\mathbf{Y} + \mathbf{v}_i)$ is the Hamming weight of $\mathbf{Y} + \mathbf{v}_i$. The unique error pattern $\mathbf{E} = \mathbf{Y} + \mathbf{v}_i$ maps transmitted vector $\mathbf{v}_i$ to the received message $\mathbf{Y}$; therefore, the likelihood $f_{H_1}(\mathbf{Y} | \mathbf{V} = \mathbf{v}_i)$ depends only on $\text{wt}(\mathbf{Y} + \mathbf{v}_i)$. As assumed $\mathbb{P}(\mathbf{V} = \mathbf{v}_i; H_1) = 1/(2^{k_1})$.

The likelihood in (3) can potentially take 2^n possible values for different $\mathbf{y}$. By the system-model, the code is linear; therefore, $\mathcal{C}_1$ and its translation $\mathbf{v}_i + \mathcal{C}_1$ are identical for any $\mathbf{v}_i \in \mathcal{C}_1$. Thus, vectors $\mathbf{y}$ which lie in the same row of standard-array have the same likelihood. The standard-array of a code is well known [1]. The sum indexed over codewords in (3) will only depend on the row of standard-array

$\mathbf{v}_0 = \vec{0}$	$\mathbf{v}_1$	$\mathbf{v}_2$	$\dots$	$\mathbf{v}_{2^{k_1}-1}$
$\mathbf{g}$	$\mathbf{g} + \mathbf{v}_1$	$\mathbf{g} + \mathbf{v}_2$	$\dots$	$\mathbf{g} + \mathbf{v}_{2^{k_1}-1}$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$

(4)

in which $\mathbf{y}$ lies and vectors in the same row of the standard-array of a binary linear code have the same likelihood.

This is because the sum in right-hand side of (3) is over all the codewords. E.g., if $\mathbf{y} = \mathbf{g} + \mathbf{v}_1$ and $\mathbf{y}' = \mathbf{g} + \mathbf{v}_2$, then the set $\{\text{wt}(\mathbf{y} + \mathbf{v}_i), \mathbf{v}_i \in \mathcal{C}_1\}$ is a permutation of $\{\text{wt}(\mathbf{y}' + \mathbf{v}_i), \mathbf{v}_i \in \mathcal{C}_1\}$. Upon addition, they both result in same value of the likelihoods, that is, $f_{H_1}(\mathbf{y}) = f_{H_1}(\mathbf{y}')$. In other words, the sum in (3) depends on the coset of $\mathbf{Y}$. Thus, if $\mathbf{Y} = \mathbf{g} + \mathbf{V}$ for some codeword $\mathbf{V} \in \mathcal{C}_1$, then

$$f_{H_1}(\mathbf{Y}) = \frac{\mathbb{P}(\mathbf{Y} \in \mathcal{C}_1 + \mathbf{g})}{2^{k_1}} = \frac{1}{2^{k_1}} \sum_{i=0}^n A_{1i}^{(\mathbf{g})} p^i (1-p)^{n-i} \quad (5)$$

where $A_{1_i}^{(\mathbf{g})}$ is the coset weight distribution of $\mathcal{C}_1 + \mathbf{g}$. Similarly,

$$f_{H_2}(\mathbf{Y}) = \frac{1}{2^{k_2}} \sum_{i=0}^n A_{2i}^{(\mathbf{h})} p^i (1-p)^{n-i} \quad (6)$$

where $\mathbf{Y}$ belongs in the row of standard-array of $\mathcal{C}_2$ with coset leader $\mathbf{h}$ and $A_{2i}^{(\mathbf{h})}$ is the weight distribution of this coset.

The likelihoods $f_{H_j}(\mathbf{Y})$, $j = 1, 2$ obtained by using (5) and (6) can be used for solving the detection problem in (2). A comment on complexity is in order. Obtaining $f_{H_j}(\mathbf{Y})$ from (5) and (6) requires the knowledge of the coset weight-distribution of the code, which is NP-hard [13]. While it seems that the situation is hopeless, it is shown in Sec. IV that if the code has a junction-tree structure or BCJR trellis structure with efficient decoding, its likelihood can be computed efficiently as well. The discussion on Chernoff information follows next.

B. Chernoff information for the code-detection problem

In hypothesis testing with repeated i.i.d. observations, the optimal error-exponent is achieved by the likelihood test and the exponent is called as Chernoff information [3, Ch. 12]. The Chernoff information between the two distributions in (5) and (6) will be inspected in this section. Let P and Q denote the probability mass functions of $f_{H_1}(\mathbf{Y})$ and $f_{H_2}(\mathbf{Y})$, respectively, corresponding to the 2^n possible values of received vector $\mathbf{Y}$. Let $P := [p_{\mathbf{y}_0}, p_{\mathbf{y}_1}, \dots, p_{\mathbf{y}_{2^n-1}}]$ and $Q := [q_{\mathbf{y}_0}, q_{\mathbf{y}_1}, \dots, q_{\mathbf{y}_{2^n-1}}]$. The optimal error exponent for the code-detection problem is given by the Chernoff information $C(P, Q)$ between the distributions P and Q [3, Ch. 12],

$$C(P, Q) = - \min_{0 \leq \lambda \leq 1} \log \left(\sum_i p_{\mathbf{y}_i}^\lambda q_{\mathbf{y}_i}^{1-\lambda} \right). \quad (7)$$

Finding the error exponent in (7) requires the knowledge of the distributions P and Q . However, obtaining the distributions is NP-hard since as discussed in Sec. III-A, computing each term in P and Q is itself NP-hard. A non-zero lower bound on $C(P, Q)$ will be developed which ensures that the detection error-probability decreases exponentially in the number of observations. For this, consider the lower-bound on $C(P, Q)$,

$$C(P, Q) \geq -\frac{1}{2} \log (1 - (d_{TV}(P, Q))^2) \quad (8)$$

where $d_{TV}(P, Q) = \frac{1}{2} \sum_i |p_{\mathbf{y}_i} - q_{\mathbf{y}_i}|$ [14]. A lower bound on $d_{TV}(P, Q)$ will be obtained which, in turn, provides a lower bound on $C(P, Q)$ from (8). For deriving lower bound on $C(P, Q)$, the following bounds on $p_{\mathbf{y}_i}$ are noted where $\mathbf{y}_i \notin \mathcal{C}_1$

$$p_{\mathbf{y}_0} \left(\frac{p}{1-p} \right)^{n-k_1} \leq p_{\mathbf{y}_i} \leq p_{\mathbf{y}_0} \frac{1 - (1-2p)^{k_1+1}}{1 + (1-2p)^{k_1+1}}, \quad (9)$$

where $p < 1/2$ [15], [16]. By (3), $2^{k_1} p_{\mathbf{y}_0}$ is equal to $\mathbb{P}(\mathbf{Y} \in \mathcal{C}_1; H_1)$. Similarly, $2^{k_1} p_{\mathbf{y}_i}$ is $\mathbb{P}(\mathbf{Y} \in \text{Coset of } \mathcal{C}_1 \text{ with } \mathbf{y}_i)$. These coset probabilities have been bounded in some classical papers [15]. In (9), denote the lower bound as $p_{\mathbf{y}_L}$ and upper bound as $p_{\mathbf{y}_H}$. Then the following main result is noted.

Theorem 1: Let $p_{\mathbf{y}_H}$ and $p_{\mathbf{y}_L}$ be the upper and lower bound on $p_{\mathbf{y}_i}$, and $q_{\mathbf{y}_H}$ and $q_{\mathbf{y}_L}$ be the upper and lower bound on $q_{\mathbf{y}_i}$ as described in (9). Without loss of generality, let $p_{\mathbf{y}_0} \geq q_{\mathbf{y}_0}$.

Let m be the dimension of the intersection subspace of $\mathcal{C}_1$ and $\mathcal{C}_2$. Then the lower bound on $d_{TV}(P, Q)$ is given by

$$d_{TV}(P, Q) \geq \frac{2^m(p_{\mathbf{y}_0} - q_{\mathbf{y}_0}) + (2^{k_1} - 2^m)(p_{\mathbf{y}_0} - q_{\mathbf{y}_H}) + (2^{k_2} - 2^m)\alpha + (2^n - 2^{k_1} - 2^{k_2} - 2^m)\beta}{2^n}$$

where $\alpha = \max\{(q_{\mathbf{y}_0} - p_{\mathbf{y}_H}), (p_{\mathbf{y}_L} - q_{\mathbf{y}_0}), 0\}$ and $\beta = \max\{(p_{\mathbf{y}_L} - q_{\mathbf{y}_H}), (q_{\mathbf{y}_L} - p_{\mathbf{y}_H}), 0\}$. Consequently, $d_{TV}(P, Q) > 0$.

Proof: Divide all possible 2^n received vectors $\mathbf{Y}$ depending on whether it belongs to code $\mathcal{C}_1 \cap \mathcal{C}_2$, $\mathcal{C}_1 \cap \mathcal{C}_2^c$, $\mathcal{C}_1^c \cap \mathcal{C}_2$, and $\mathcal{C}_1^c \cap \mathcal{C}_2^c$. The cardinalities of these sets are 2^m , $2^{k_1} - 2^m$, $2^{k_2} - 2^m$, and $2^n - 2^{k_1} - 2^{k_2} - 2^m$, respectively.

Since $d_{TV}(P, Q) = \frac{1}{2} \sum_i |p_{\mathbf{y}_i} - q_{\mathbf{y}_i}|$, so it is required to bound $|p_{\mathbf{y}_i} - q_{\mathbf{y}_i}|$ on these four sets.

For $\mathbf{y}_i \in \mathcal{C}_1 \cap \mathcal{C}_2$, the difference is simply $|p_{\mathbf{y}_i} - q_{\mathbf{y}_i}| = |p_{\mathbf{y}_0} - q_{\mathbf{y}_0}| = p_{\mathbf{y}_0} - q_{\mathbf{y}_0}$ by the assumption $p_{\mathbf{y}_0} \geq q_{\mathbf{y}_0}$.

For $\mathbf{y}_i \in \mathcal{C}_1 \cap \mathcal{C}_2^c$ the difference is $|p_{\mathbf{y}_i} - q_{\mathbf{y}_i}| = |p_{\mathbf{y}_0} - q_{\mathbf{y}_i}| = p_{\mathbf{y}_0} - q_{\mathbf{y}_i} \geq p_{\mathbf{y}_0} - q_{\mathbf{y}_H}$ where we used $p_{\mathbf{y}_0} > q_{\mathbf{y}_0} > q_{\mathbf{y}_H} > q_{\mathbf{y}_i}$ for the inequality.

For $\mathbf{y}_i \in \mathcal{C}_1^c \cap \mathcal{C}_2$ the difference is $|p_{\mathbf{y}_i} - q_{\mathbf{y}_i}| = |p_{\mathbf{y}_i} - q_{\mathbf{y}_0}|$. This difference is obviously ≥ 0 . Further, $|p_{\mathbf{y}_i} - q_{\mathbf{y}_0}| \geq (p_{\mathbf{y}_i} - q_{\mathbf{y}_0}) \geq (p_{\mathbf{y}_L} - q_{\mathbf{y}_0})$. And $|p_{\mathbf{y}_i} - q_{\mathbf{y}_0}| \geq (-p_{\mathbf{y}_i} + q_{\mathbf{y}_0}) \geq (-p_{\mathbf{y}_H} + q_{\mathbf{y}_0})$. Thus, we get $|p_{\mathbf{y}_i} - q_{\mathbf{y}_0}| \geq \alpha$.

For $\mathbf{y}_i \in \mathcal{C}_1^c \cap \mathcal{C}_2^c$ the difference is $|p_{\mathbf{y}_i} - q_{\mathbf{y}_i}| = |p_{\mathbf{y}_i} - q_{\mathbf{y}_i}|$. A similar procedure (as in previous paragraph) results in $|p_{\mathbf{y}_i} - q_{\mathbf{y}_i}| \geq \beta$.

Putting together the cardinalities of $\mathcal{C}_1 \cap \mathcal{C}_2$, $\mathcal{C}_1 \cap \mathcal{C}_2^c$, $\mathcal{C}_1^c \cap \mathcal{C}_2$, and $\mathcal{C}_1^c \cap \mathcal{C}_2^c$ with the bounds on $|p_{\mathbf{y}_i} - q_{\mathbf{y}_i}|$ we get the result. If $\mathbf{y}_i \in \mathcal{C}_1$, $\mathbf{y}_i \notin \mathcal{C}_2$, $p_{\mathbf{y}_0} - q_{\mathbf{y}_H} > 0$. Thus, $d_{TV}(P, Q) > 0$. ■ By above theorem, the Chernoff information for code-detection problem is positive; so, the code-detection error probability decays exponentially with multiple observations.

IV. FAST CALCULATION METHODS FOR $f_{H_j}(\mathbf{Y})$

It was noted that calculation of $f_{H_j}(\mathbf{y})$, $j = 1, 2$ is NP-hard in Sec. III-A. For simulations, the calculation required in (3) is challenging. In this section, we leverage on efficient decoding algorithms, when available, for efficient computation of the likelihood $f_{H_j}(\mathbf{y})$, $j = 1, 2$. In particular, if the two hypotheses $\mathcal{C}_1$ and $\mathcal{C}_2$ have efficient junction tree structures, then generalized distributive law (GDL) techniques [4] can be used to compute the likelihood efficiently. Similarly, if efficient BCJR trellis can be associated with the two channel codes [5], then the likelihood $f_{H_j}(\mathbf{y})$ can be computed efficiently.

A. Likelihood calculations based on the GDL

If a channel-code admits a junction-tree structure, the GDL technique can be used to decode a bit in a maximum likelihood fashion [4]. The GDL is a message passing algorithm that provides an efficient method to compute functions which can be written as marginalization over a product of several functions [4]. When channel errors are i.i.d. the likelihood $f_{H_1}(\mathbf{Y}|V_i = v_i)$ can be computed efficiently using GDL. Having obtained $f_{H_1}(\mathbf{Y}|V_i = 0)$ and $f_{H_1}(\mathbf{Y}|V_i = 1)$ by GDL, a method to compute the likelihoods $f_{H_1}(\mathbf{Y})$ and $f_{H_2}(\mathbf{Y})$ will be shown next.

First note that the conditional probability $\mathbb{P}(V_i = 1|\mathbf{Y}; H_1)$ of i -th bit V_i of the vector $\mathbf{V}$ can be written as

$$\begin{aligned} \mathbb{P}(V_i = 1|\mathbf{Y}; H_1) &= \sum_{v_i=1, \mathbf{v} \in \mathcal{C}_1} \mathbb{P}(\mathbf{V} = \mathbf{v}|\mathbf{Y}; H_1) \\ &= \sum_{v_i=1, \mathbf{v} \in \mathcal{C}_1} \frac{\mathbb{P}(\mathbf{V} = \mathbf{v})}{f_{H_1}(\mathbf{Y})} f_{H_1}(\mathbf{Y}|\mathbf{V} = \mathbf{v}) \\ &= \frac{1}{2^k} \frac{1}{f_{H_1}(\mathbf{Y})} \sum_{v_i=1, \mathbf{v} \in \mathcal{C}_1} f_{H_1}(\mathbf{Y}|\mathbf{V} = \mathbf{v}) \\ &= \frac{1}{2^k} \frac{1}{f_{H_1}(\mathbf{Y})} L_{H_1}(1), \end{aligned} \quad (10)$$

where $\mathbb{P}(\mathbf{V} = \mathbf{v}) = (1/2^{k_1})$ and $f_{H_1}(\mathbf{Y})$ does not depend on the vector $\mathbf{v}$. The function $L_{H_1}(1)$ is a brief notation for $\sum_{v_i=1, \mathbf{v} \in \mathcal{C}_1} f_{H_1}(\mathbf{Y}|\mathbf{V} = \mathbf{v})$. Similar expressions can be derived for $v_i = 0$ and for the alternate hypothesis H_2 . Since $\mathbb{P}(V_i = 1|\mathbf{Y}; H_1) + \mathbb{P}(V_i = 0|\mathbf{Y}; H_1) = 1$, therefore,

$$\frac{1}{2^k} \frac{1}{f_{H_1}(\mathbf{Y})} [L_{H_1}(0) + L_{H_1}(1)] = 1. \quad (11)$$

Next, by using i.i.d. nature of the error bits, it will be shown that $L_{H_1}(0), L_{H_1}(1)$ can be computed by evaluating a sum-of-product function, which is addressed by GDL.

The functions $L_{H_1}(0)$ and $L_{H_1}(1)$ can be written as sum-of-product as shown by Aji and McEliece [4, Ex. 2.3]. In linear block codes, any $\mathbf{V} \in \mathcal{C}_1$ satisfies $(n - k_1)$ parity check equations. Define an indicator random variable $I_j^0(\mathbf{V})$ as

$$I_j^0(\mathbf{V}) = \begin{cases} 1 & \text{if } j\text{th parity check equation of the code} \\ & \text{is satisfied with } V_i = 0 \\ 0 & \text{otherwise} \end{cases}$$

Since the received vector $\mathbf{Y}$ is the addition of i.i.d. noise to the transmitted codeword, $L_{H_1}(0)$ can be written as,

$$L_{H_1}(0) = \sum_{v_i=0, \mathbf{v} \in \mathcal{C}_1} \prod_{i=1}^n \mathbb{P}(Y_i|V_i = v_i; H_1) \prod_{j=1}^{n-k_1} I_j^0(\mathbf{v}), \quad (12)$$

where the indicator represents the parity-checks. The right hand side of (12) is in the sum-of-product form. If the channel-code $\mathcal{C}_1$ has a junction tree structure for efficient decoding, then the same structure can be utilized to compute $L_{H_1}(0)$ and $L_{H_1}(1)$ efficiently. Similarly, $L_{H_2}(0)$ and $L_{H_2}(1)$ can be computed. Methods to obtain junction trees are known [17].

B. Likelihood calculations based on the BCJR algorithm

A linear block code can be represented by a trellis and the BCJR algorithm is a method to perform maximum-likelihood bit-decoding. That is, given a received vector $\mathbf{Y} = \mathbf{y}$, the BCJR algorithm will maximize $\mathbb{P}(V_i = v_i|\mathbf{Y} = \mathbf{y}; H_1)$ for each i [5]. Efficient techniques to represent any linear block codes using a trellis are given in [18]. The desired likelihoods $f_{H_j}(\mathbf{Y}), j = 1, 2$ can be found in intermediate steps of the BCJR algorithm. This is outlined next.

An example of BCJR trellis corresponding to some linear block code is shown in Fig. 2 [18]. A trellis consists of nodes which can be in any of M distinct states and edges which

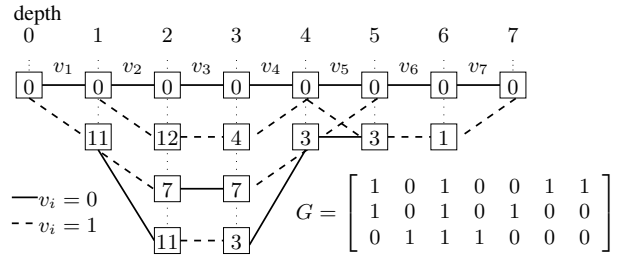


Fig. 2. Trellis for linear block code defined by G is illustrated [18]. The paths from depth 0 to depth 7 represent the codewords. Each path visits states at various depths, for which likelihood can be computed.

represent bits of a codeword. A sequence of edges connecting the start node and the terminal node correspond to a codeword $\mathbf{v} = [v_1 \cdots v_n]$, and all possible codewords are represented by all possible sequences of edges. Let S_i denote the state at depth i , for $i = 0$ to n . For example in Fig. 2, S_1 can be 0 or 11 depending on which codeword is chosen. Let $\lambda_i(m) = \mathbb{P}(S_i = m|\mathbf{Y}; H_1)$. The BCJR algorithm calculates $\lambda_i(m)$ in an intermediate step. The desired likelihood can then be obtained as $f_{H_1}(\mathbf{Y}) = \sum_{m=0}^{M-1} \lambda_i(m)$. Similarly, the likelihood $f_{H_2}(\mathbf{Y})$ can be obtained. The BCJR algorithm is applicable to linear codes which can be represented by a trellis. The BCJR algorithm is a special case of the GDL algorithm [4]; but, the GDL algorithm and the BCJR algorithms require the junction tree and the trellis of the code, respectively.

V. COMPARISON OF LIKELIHOOD RATIO TESTS

Following simulations results will be presented next. First, the Chernoff information will be compared against the derived lower bound in Theorem 1. Second, in the literature, a method based on inner-product with row of parity-check matrix has been used to detect $H_1 : \mathcal{C}_1$ against a null-hypothesis H_0 where data is Bernoulli(1/2) [7]. Similarly, convolutional code detection in a hypothesis testing setup has been addressed by examining the parity checks [6]. Their methods, based on parity-check or inner-product with row of parity-check matrix, is adapted to our code-detection problem and their Chernoff informations are compared against the Chernoff information for our likelihood test. Third, the average error probability in the code-detection problem is compared for the likelihood test against the methods available in literature.

Two hypothesis testing problems are considered for simulations. In first H_1 : Hamming(15,11) and H_2 : BCH(15,7), while in second H_1 : Hamming(31,26) and H_2 : BCH(31,16). The results are plotted in Fig. 3 and the inner-product method is explained next.

A vector $\mathbf{s} \in \mathcal{C}_1^\perp, \mathbf{s} \notin \mathcal{C}_2^\perp$ is chosen with smallest possible weight. From received sequence $\mathbf{Y}_1^N$, the inner product $\langle \mathbf{s}, \mathbf{Y}_i \rangle$ is obtained. The distribution of $\langle \mathbf{s}, \mathbf{Y}_i \rangle$ is expected to be [7]:

$$H_1 : \langle \mathbf{s}, \mathbf{Y}_i \rangle \sim \text{Bernoulli} \left(\frac{1 - (1 - 2p)^{\text{wt}(\mathbf{s})}}{2} \right)$$

$$H_2 : \langle \mathbf{s}, \mathbf{Y}_i \rangle \sim \text{Bernoulli}(1/2)$$

The Chernoff information or error-probability corresponding to these distributions is labeled as inner-product method.

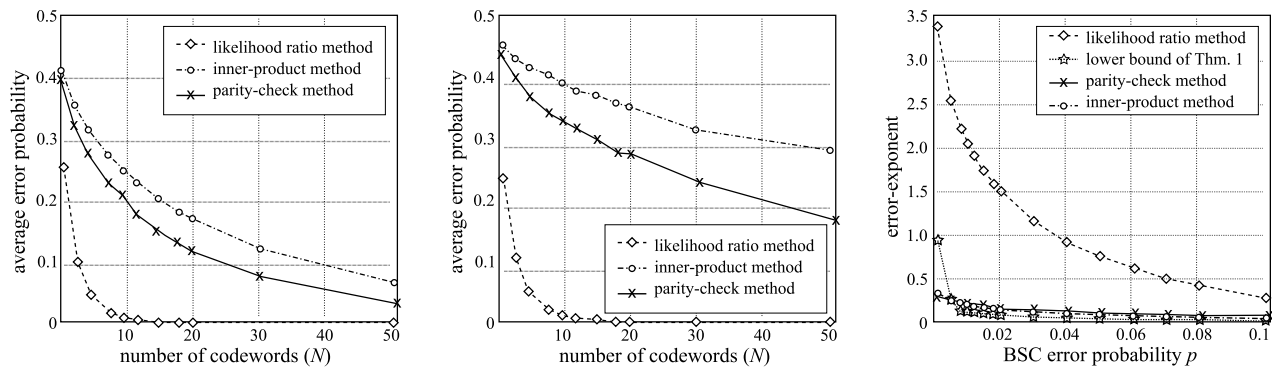


Fig. 3. (a) The two hypotheses are H_1 : Hamming(15,11) and H_2 : BCH(15,7), and the average error probability is plotted with $p = 0.1$. (b) The two hypotheses are H_1 : Hamming(31,26) and H_2 : BCH(31,16), and the average error probability is plotted with $p = 0.1$. (c) The two hypotheses are H_1 : Hamming(15,11) and H_2 : BCH(15,7), and the Chernoff information is compared for different tests. Overall, the likelihood test performs much better than the other two tests. For the parity-check method, the Chernoff information was obtained empirically (by simulations).

Code-detection problem with two convolutional codes as hypotheses has been studied by utilizing parity-check equations [6]. A parity-check equation $s \in \mathcal{C}_1^\perp, s \notin \mathcal{C}_2^\perp$ must be satisfied by all codewords under hypothesis H_1 but not under H_2 . This idea based on parity-checks is adapted to linear block codes, and is included for comparisons. In [6], additive white Gaussian noise channel with binary phase shift keying modulation is considered. An equivalent signal to noise ratio is used to match the error probability p for comparative simulations [3, Ch. 9].

The empirical average error probability for our method is compared against the two methods adapted from the literature to our code-detection problem. It is observed that likelihood tests outperform the other tests for small block-length codes. We expect similar results for larger block-lengths as well, since likelihood tests are *fundamentally optimal* for minimizing error-probability. Simulations for larger block-length are desired, but as noted in Sec. III-A complexity of likelihood computation scales rapidly as $\mathbf{Y}$ can take 2^n distinct values.

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VII. CONCLUSIONS AND FUTURE WORK

This work introduced and studied a code-detection problem of interest in the cognitive paradigm or for security applications. By observing a large number of noise affected codewords, a third-party receiver has to detect the channel-code being used by the transmitter. *Fundamentally optimal* likelihood ratio tests were used for minimizing the error probability in this detection problem. It was shown that the Chernoff information for this detection problem is strictly positive, that is the error probability decreases exponentially with number of repeated observations. Further, the likelihood test (as expected) performed better than parity-check based schemes of detection. Complexity in computing likelihood for a general binary linear block code is an NP-hard problem. This was addressed using the generalized distributive law techniques or BCJR algorithm.

These techniques are applicable when the channel-code itself is amenable to efficient decoding.

Likelihood ratio test for multiple hypotheses and unequal block-length of the two channel-codes is of interest. We also wish to apply likelihood test for convolutional code-detection.

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